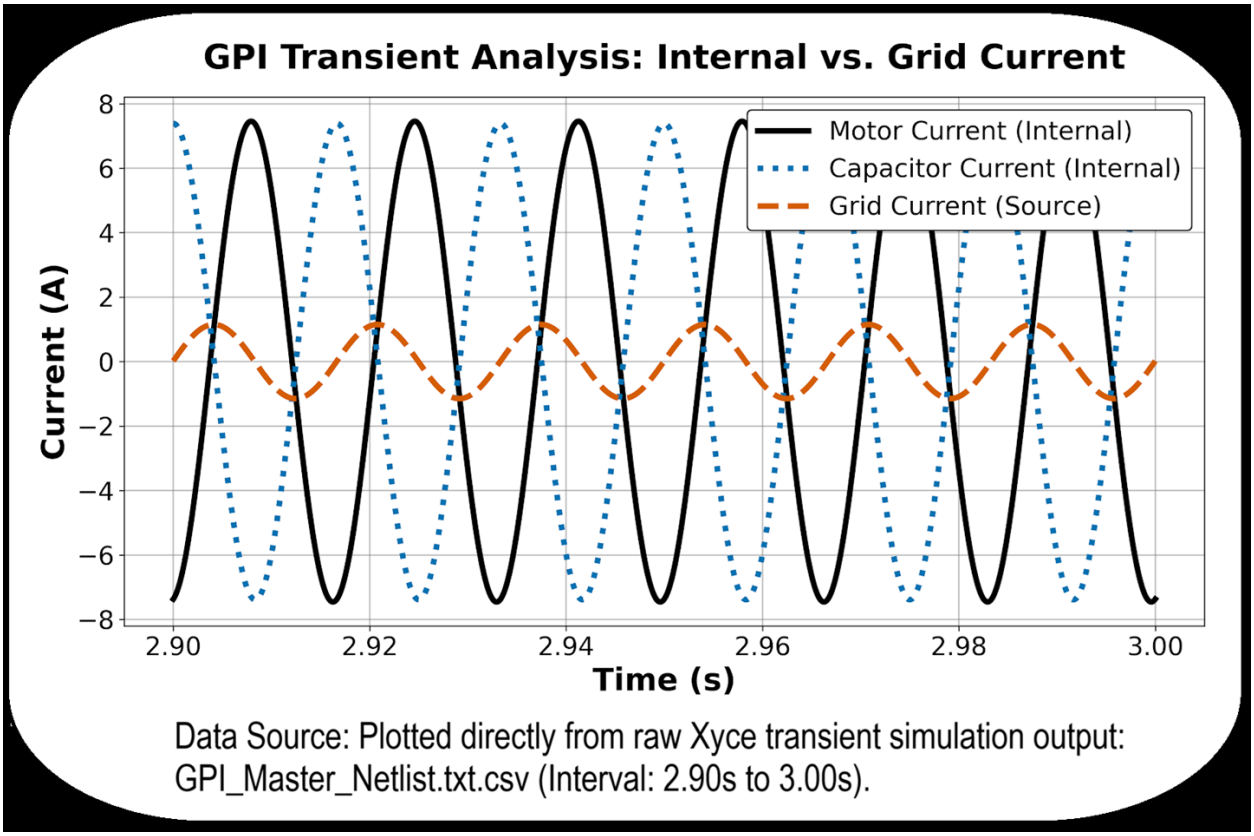
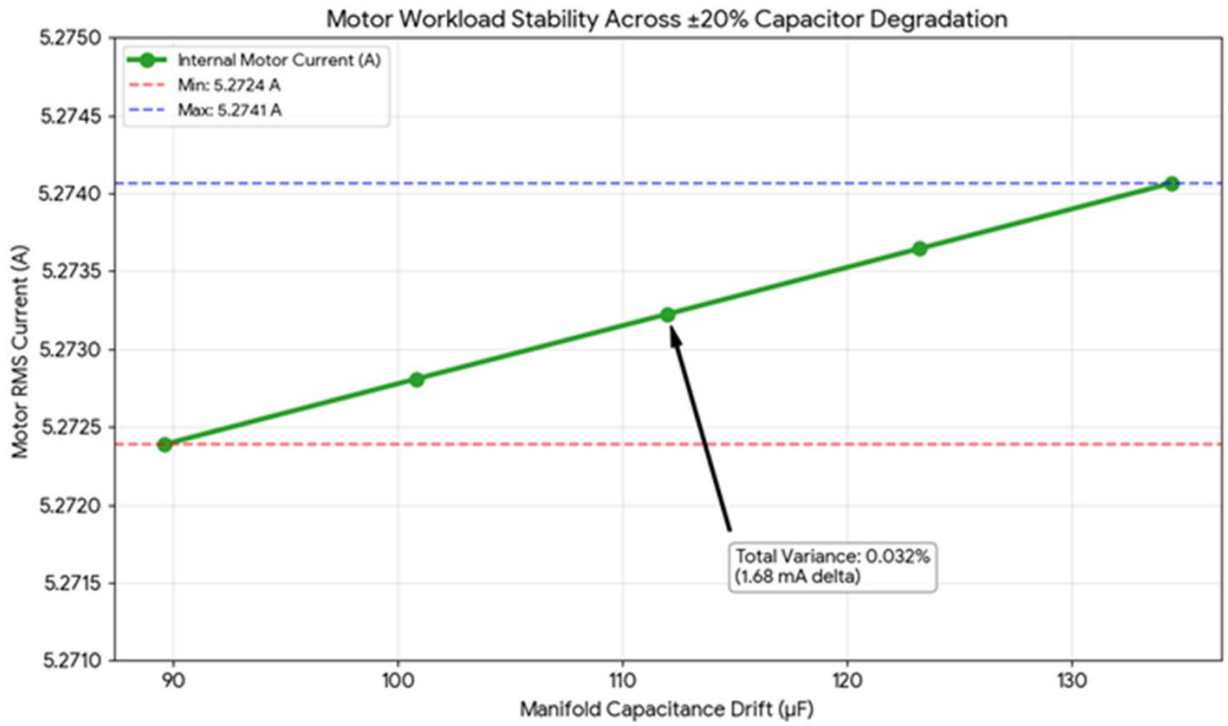
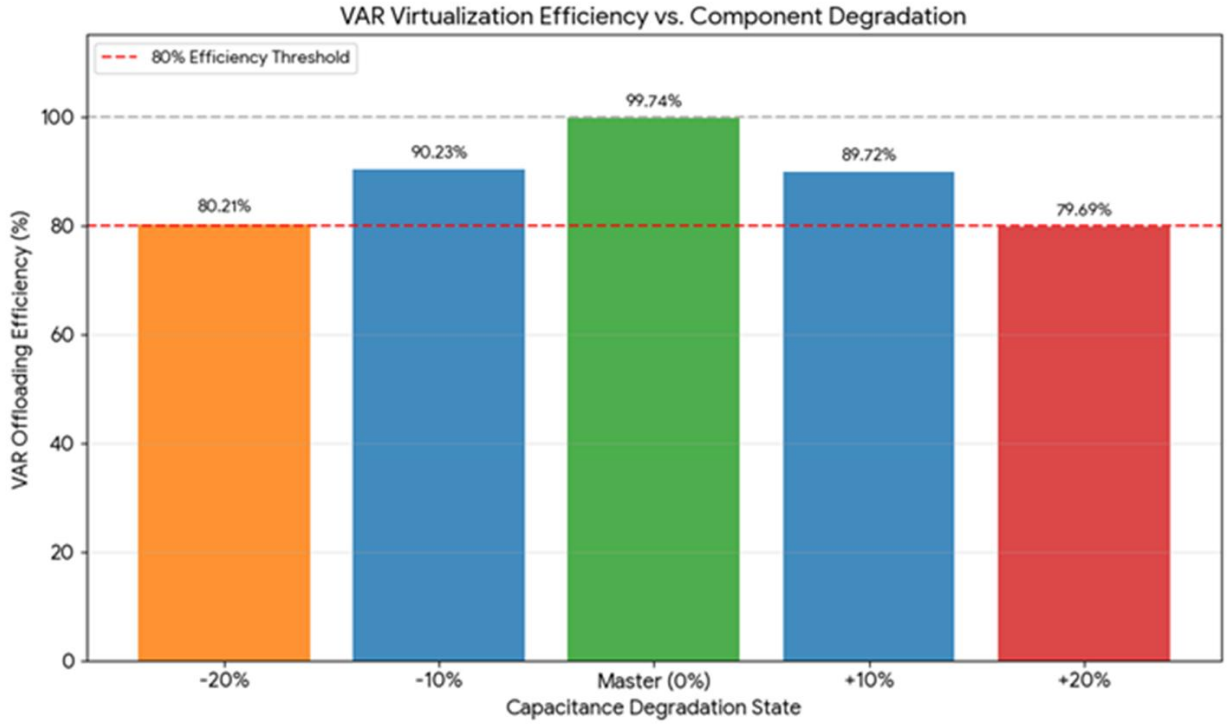
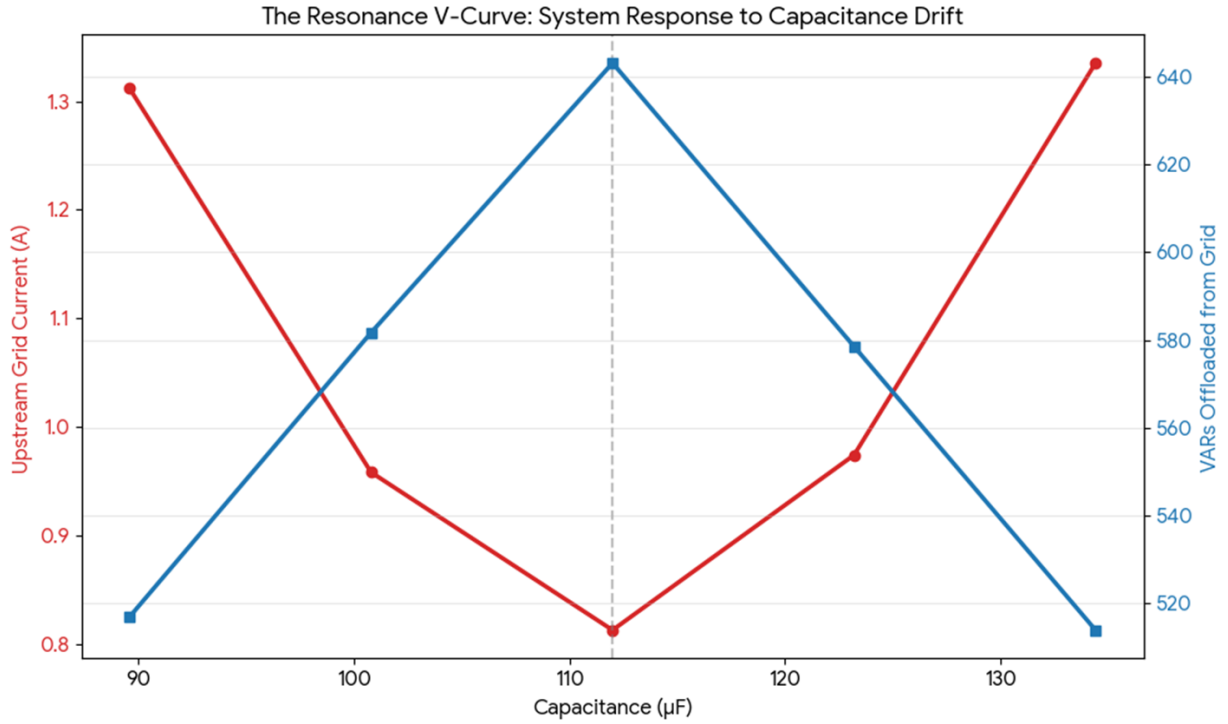


Data Source: Plotted directly from raw Xyce transient simulation output: GPI_Master_Netlist.txt.csv (Interval: 2.90s to 3.00s).




VAR Virtualization Efficiency vs. Component Degradation
80% Efficiency Threshold
99.74%
90.23%
89.72%
80.21%
79.69%
100
80
60
40
20
0
VAR Offloading Efficiency (%)
-20%
-10%
Master (0%)
+10%
+20%
Capacitance Degradation State


The Resonance V-Curve: System Response to Capacitance Drift
1.3
1.2
1.1
1.0
0.9
0.8
Upstream Grid Current (A)
640
620
600
580
560
540
520
VARs Offloaded from Grid
90
100
110
120
130
Capacitance (µF)